

LoomLedger — Project Documentation

Authentic Indian Handloom Platform — Decentralized Trust & Direct Marketplace

1. Executive Summary & Core Vision

LoomLedger is an end-to-end digital trust platform and direct artisan marketplace designed to preserve India's rich handloom heritage. By combining **Polygon Blockchain Digital Product Passports (DPP)** with modern Web technologies, LoomLedger protects authentic handloom weavers from power-loom counterfeits, ensures verifiable Geographical Indication (GI) provenance, and eliminates exploitative intermediaries.

- **Verifiable Authenticity:** Issue tamper-proof, on-chain Digital Passports for authentic handlooms.
- **Artisan Empowerment:** Enable certified weavers to list products directly and receive 100% fair payout.
- **Consumer Trust:** Instant QR-code scanning revealing raw materials, weaver identity, and craft duration.

2. System Architecture & Tech Stack

Layer	Technologies Used	Key Responsibilities
Frontend	Next.js 15 (App Router), React 19, Tailwind CSS, Framer Motion	User interface, Toast notifications, Target price selector, Buyer settings, Admin console
Backend API	Node.js, Express.js, TypeScript, Rate Limiting, Helmet	REST endpoints, JWT authentication, Cloudinary uploads, URL rewrite, Cascade deletions
Database	PostgreSQL (Neon Cloud), Prisma ORM	Relational schema, serverless connection pooler, user & product data
Blockchain	Solidity, Hardhat, Ethers.js, Polygon Amoy Testnet	ERC-721 Digital Product Passports, Artisan Verification Registry

3. Database Schema (Prisma ORM & PostgreSQL)

- **User Model:** Handles credentials, JWT auth, roles (buyer, artisan, admin), and profile info.
- **Artisan Model:** Stores craft specialization, region, GI certificate number, verification status, and wallet address.
- **Product Model:** Marketplace inventory including price, fabric, craft time, categories, stock, and blockchain token ID.
- **DigitalPassport Model:** Stores Polygon NFT metadata, token ID, transaction hash, raw material source, and weaver ID.
- **Order & OrderItem Models:** Tracks purchase history, payment status, and saved recipient shipping addresses.

4. Smart Contract Architecture (Polygon Amoy)

1. **ArtisanRegistry.sol:** Registers and verifies GI-certified master weavers on-chain.
2. **PassportContract.sol (ERC-721):** Mints immutable Digital Product Passports containing weave duration and craft lineage.

5. Key REST API Endpoints Specification

Method	Endpoint	Access	Description
POST	/api/auth/login	Public	Authenticates user credentials and returns JWT token
GET	/api/artisans	Public	Lists approved master weavers (query ?status=all)
GET	/api/artisans/:id	Public	Fetches artisan profile by artisanId or userId
DELETE	/api/admin/artisans/:id	Admin	Deletes artisan account and purges ALL listed products
GET	/api/products	Public	Fetches active marketplace product inventory
POST	/api/orders	Buyer	Places a new order with auto-filled shipping address

6. Key UX Innovations & Frontend Components

- **Admin Artisan Purge & Product Cascade:** Admin console allows single-click artisan deletion which cascades to purge all listed marketplace products, cart items, and passports.
- **Interactive Target Price Selector:** 1-click price preset chips (■1k, ■2.5k, ■5k, ■10k) and ±■100, ±■500 step buttons.
- **Buyer Profile & Shipping Address:** Dedicated dashboard (/buyer/settings) allowing buyers to manage default delivery addresses with automatic checkout pre-fill.
- **Toast Notification System:** Global animated toast provider with Log In / Sign Up action prompts replacing browser alert() popups.

7. Security, CORS & Reliability Specifications

1. **Dynamic CORS Whitelisting:** Configured to allow all Vercel deployment origins (*.vercel.app) and local dev.
2. **URL Auto-Rewrite Middleware:** Automatically redirects requests missing the /api prefix to prevent 404 errors.
3. **Neon DB Connection Pool Timeout:** Appends connect_timeout=30 and pool_timeout=30 to handle serverless idle dropoffs.

8. Deployment Details & Live Links

- **Frontend Deployment (Vercel):** <https://loom-ledger.vercel.app>
- **Backend API Server (Render):** <https://loomledger-d1zj.onrender.com/api>
- **Polygon Amoy Explorer:**
<https://amoy.polygonscan.com/address/0x551049E4bb3eB377db9e24B65b83b226fc3Dc1eA>